

Gonçalo Pereira

Network Systems · Linux Kernel Internals · Backend Development

goncalo.pereira.1021@gmail.com github.com/goncalo1021pt goncalo.fontao.net LinkedIn +351 925 526 272 Lisbon, PT
Systems developer working in C on the packet path of telecom gateways — raw-socket diagnostics, kernel modules in the receive path, netlink routing. Also ships and runs a production Go service on a home server.

EXPERIENCE

Embedded Software Engineer · Streambow

Dec 2025 – Present · Sintra, Portugal

- Develop the network diagnostics agent shipped on telecom operator gateways: raw-socket ICMP, TWAMP, throughput and HTTP browsing probes reporting over MQTT
- Added a traceroute diagnostic measuring per-hop RTT and loss, so support can locate *where* a subscriber path degrades instead of only that it does
- Diagnosed and fixed two gateway crash mechanisms in the receive path: a per-CPU backlog fallback that re-injected every packet under RPS until the watchdog fired, and a double free on upload
- Replaced brittle route parsing with libnl queries against the kernel route cache, and made WAN detection event-driven by subscribing to netlink route notifications
- Ported the agent and its three kernel modules onto seven OEM gateway platforms across prpIOS, OpenWrt and buildroot — ARMv7 and aarch64, musl and glibc, kernels up to 5.4, GCC 5.5 to 7.5
- Automated customer releases end to end — compile, strip, document, checksum and package in one command — and publish container targets as versioned OCI images

Student Supervisor · 42 Lisboa

Jul 2024 – Oct 2024 · Lisbon, Portugal

- Reviewed and evaluated C and systems programming projects against specification
- Delivered technical presentations to audiences of 100+ students

PROJECTS

Quest Board · dnd.fontao.net · Go · PostgreSQL · React · OpenAPI · Docker

2026

- Designed, shipped and operate a production web app on my own server: a React SPA in a single Go binary
- Contract-first API — one OpenAPI spec generates the Go server interface and the typed TS client
- OAuth2, local accounts and TOTP 2FA; CI-gated merges, staging and production deploy pipelines

Homelab Observability · Prometheus · Grafana · Docker Compose

2026

- Metrics for Quest Board, its host and every container in the homelab, on 30-day retention
- Dashboards and alert rules provisioned as code — a new service ships already monitored
- Grafana published through Cloudflare Access; Prometheus and exporters never publicly exposed

Dune Engine · C++ · 45k lines · Systems Architecture

2026 · in progress

- Turn-based board-game engine: phase state machine, event bus, and adapters for TTY and scripted play

42 Lisboa Projects · C · Go · x86-64 Assembly · Rust

2023 – 2026

- **Networking** — ping and traceroute rebuilt on raw sockets: handcrafted ICMP, TTL control, RTT timing
- **Systems** — a thread-safe malloc over mmap; a POSIX shell driving fork/exec, pipes, redirections and job signals from an AST; a C standard library and x86-64 assembly primitives from scratch
- **Security** — an ARP-spoofing MITM inspector and a TOTP generator in Go, a Tor hidden service, binary reverse engineering, and an educational ransomware simulation in Rust

EDUCATION

42 Lisboa · Systems & Kernel Specialisation · RNCP Level 7 (Master's equivalent)

Oct 2023 – Present · Lisbon, Portugal

- Common core followed by a post-core specialisation, both assessed entirely through projects
- Network protocol implementation, memory allocators, x86-64 assembly and a cybersecurity track

SKILLS

Languages	C, C++, Go, Python, Bash, x86-64 Assembly, SQL, TypeScript, Rust (basics)
Networking	Raw sockets, TCP/IP, ICMP, netlink/libnl, MQTT, packet analysis (Wireshark, tcpdump)
Linux	Kernel modules, network stack, syscalls, signals, threads, memory management, cross-compilation
Observability	Prometheus, Grafana, metrics instrumentation, alerting as code
Backend	Go (chi, sqlc, pgx), PostgreSQL, REST and OpenAPI, OAuth2/JWT/TOTP
Infrastructure	Docker, Docker Compose, GitHub Actions CI/CD, NGINX, Make, Git, GDB, Valgrind

Portuguese (Native) · English (Fluent)